ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CAS5511 ARTIFICIAL INTELLIGENCE

DEFERED EXAMINATION

DATE: 18-06-2025

TIME: 14:00HRS – 17:00HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted (Practical, Scenario based or Problem based)

Question 1

(Total: 50 marks)

If you ask about artificial intelligence to an AI researcher, (s)he would say that it's a set of algorithms that can produce results without having to be explicitly instructed to do so. The intelligence demonstrated by machines is known as Artificial Intelligence. Artificial Intelligence has grown to be very popular in today's world. It is the simulation of natural intelligence in machines that are programmed to learn and mimic the actions of humans. These machines are able to learn with experience and perform human-like tasks. As technologies such as AI continue to grow, they will have a great impact on our quality of life.

a) State and briefly describe 4 tests that help to determine the 'Human Likeness' of artificial intelligence systems (40 marks)
b) Describe any 2 sub domains of artificial intelligence (10 marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

(25 marks)

The theory of Artificial Intelligence predicts that with time it will transition through 3 main stages.

a) Describe each of these transition stages giving key characteristic differences of each (15 marks)
b) What is the difference between Deep Learning and Machine Learning? (10 marks)

Question 3

(25 marks)

Artificial Intelligence is revolutionizing the way in which different companies across compete and grow across the world by representing a new production factor that can drive business profitability. Big data science and increased processing capacity has increased use cases for interoperability of different systems making advanced systems simpler to use

a) Describe 4 ways in which AI has turned research and development into a needed business asset for many organisations (20 marks)
b) State any 5 real world implementation possibilities of Artificial Intelligence today (5 marks)

Question 4

(25 marks)

In artificial intelligence, problems can be solved by using searching algorithms, evolutionary computations, knowledge representations, etc.

a) State and describe the 4 key properties of Search Algorithms (20 marks)
b) State any 5 examples of uninformed search algorithms (5 marks)

END OF EXAMINATION